Methods: NGC4429

ALMA Thesis Planner (automated pipeline)

Methods

0. Software environment and project scaffolding (week 1). Set up a single, version-controlled Python environment for the whole thesis: ‘astropy‘ (FITS I/O, WCS, units, constants), ‘spectral-cube‘ and ‘radio-beam‘ (cube handling and beam objects), ‘numpy‘/‘scipy‘ (numerics, convolution, statistics), ‘photutils‘ (aperture photometry), and ‘matplotlib‘ (all figures). Every number that appears in the thesis must be produced by a script in the repository, not by an interactive session you can't reproduce. Create one module for cube utilities, one for photometry, one for the toy-model convolution machinery, and a ‘results/‘ directory where each script writes both figures and a machine-readable table of the quantities it measured. You have exactly two input files — the primary-beam-corrected Band 3 continuum image and the 126-channel HCN(1-0) cube — and everything below operates on those alone.

1. Data verification and noise characterization (weeks 1–2). Before any science measurement, verify the data are what the headers claim:

- *Header audit.* Read both FITS headers and record, in a table that goes into the thesis: synthesized beam (BMAJ/BMIN/BPA — expect ~$1.63''\times1.55''$ for the continuum, ~$1.88''\times1.81''$ for the cube, PA ~$80°$), pixel scale, spectral axis convention (frequency vs. velocity, rest frame, channel width ~14.65 MHz $\approx$ 49.6 km/s), bunit, and reference pixel/coordinates. Convert the spectral axis to velocity relative to the HCN(1-0) rest frequency of 88.6316 GHz using the radio convention, and confirm the redshifted line for NGC 4429's systemic velocity falls well inside the 126-channel window (compute which channels those are and record them).
- *Astrometry check.* Confirm the phase/field center in the headers is consistent with the ICRS position RA = 12:27:26.503, Dec = +11:06:27.58, and locate that position's pixel coordinates in both images. All apertures below are anchored to this position, not to a fitted centroid, unless the continuum detection (Section 5) provides a better nuclear position — in which case state that and use it consistently.
- *Noise measurement.* These are primary-beam-corrected products, so the noise rises toward the field edge. Measure RMS in the inner region only (say, the central ~$30''$ where the PB correction is mild), using emission-free channels for the cube and emission-free sky regions for the continuum. For the cube, compute the per-channel RMS across all 126 channels, plot RMS versus channel, and use robust statistics (median absolute deviation scaled to σ) so any real line emission doesn't bias the estimate. Record the median per-channel RMS in mJy/beam; this single number drives every sensitivity calculation in the thesis.
- *Integrated spectrum.* Extract the spatially integrated spectrum in a circular aperture centered on the nucleus, sized to cover the known circumnuclear disk with margin (a few beams; try radii of ~$3''$, $5''$, and $10''$ and keep all three). Correctly convert Jy/beam to Jy by dividing

by the number of pixels per beam. Propagate the per-channel noise into an uncertainty per spectral channel for each aperture, accounting for the number of independent beams in the aperture.

2. The detectability floor: beam-convolution sanity checks (week 2, ~3 days). Before the gate decision, build the toy machinery you will reuse in Section 6: generate simple model disks on the cube's pixel grid — a filled exponential disk and the same disk with a central cavity of a given radius and depth — convolve each with the actual synthesized beam (use 'radio-beam' to construct the Gaussian kernel from the header values, 'scipy' or 'astropy.convolution' for the convolution), and compare the central-beam flux of holed versus filled models. This tells you, quantitatively, how large and how deep a cavity must be to produce a measurable central deficit at 1.9″ resolution, given your measured RMS. Write this up as one appendix figure now; it also serves as a check that your convolution code conserves flux (verify total model flux is unchanged by convolution to <1%).

3. The week-2 gate. Apply the pre-registered detection criterion to the integrated spectrum and the cube: the branch is *detection* if either (a) the integrated line flux over the expected velocity window is $\geq 5\sigma$, or (b) any spatial position shows per-beam peak S/N ≥ 3 in ≥ 3 contiguous channels. Compute both tests with a script, record the numbers, and write a one-paragraph gate memo (date, numbers, decision) that goes verbatim into the thesis. From this point you build tooling for exactly one branch — Section 4A or 4B, never both.

4A. Detection branch: spatial and spectral characterization.

- *Line window and masking.* Define the line channel range from the integrated spectrum (channels with signal above ~1σ, extended by one channel each side). For the moment-0 map, use a simple threshold mask: include voxels above ~2σ in at least one channel, after optionally smoothing spatially by half a beam for mask construction only (measure fluxes from the unsmoothed cube). At ~49.6 km/s channels the line likely spans only a handful of channels, so keep masking simple and document the exact recipe.

- *Moment maps.* Compute moment 0 (integrated intensity, Jy/beam km/s) over the masked window, with an uncertainty map from σ_chan × Δv × $\sqrt{}$N_chan per pixel. Compute moment 1 (intensity-weighted velocity) only if the line spans $\geq$4–5 channels across the disk; if it doesn't, report a single flux-weighted velocity centroid from the integrated spectrum instead and say so. Do not compute moment 2 — at this channel width it is dominated by channelization, and the thesis states that in one sentence rather than presenting an unreliable map.

- *Global quantities.* Report total HCN(1-0) flux (Jy km/s) from the moment-0 map within the aperture that converges (check that flux versus aperture radius flattens), the FWZI/FWHM line width from the integrated spectrum (with the ±half-channel quantization uncertainty stated), and the systemic velocity from the line centroid. Flux uncertainty combines the statistical error with a standard 5–10% ALMA Band 3 flux-calibration term, added in quadrature and stated separately.

- *Radial profile.* Azimuthally averaged moment-0 profile in half-beam-width annuli centered on the nucleus, with uncertainties from the noise map and the number of independent beams per annulus. This is the observed profile the Section 6 models are compared against.

- *Position–velocity diagram.* Extract a PV slice along the disk major axis (take the PA from the moment-1 map if you made one; otherwise from the observed elongation of the moment-0 map, and record which). Present it with the beam and channel width drawn on the panel, and use it for exactly one purpose: verifying the emission shows the velocity gradient expected of a rotating disk. No kinematic fitting, no rotation-curve extraction, no SMBH mass modeling — the resolution and channelization do not support it and the thesis says so once, plainly.

4B. Non-detection branch: upper limits and stacking.

- *Per-beam and per-aperture upper limits.* From the per-channel RMS, compute the 3σ upper limit on integrated line flux per synthesized beam for an assumed line width (compute for 50, 100, and 200 km/s — i.e., 1, 2, and 4 channels — and tabulate all three; the scaling is σ_chan Δv $\sqrt{}$N_chan). Produce a map of the per-pixel 3σ limit across the inner disk (it varies with the PB correction). Then compute aperture-integrated limits for three fixed regions: the central beam, an "inner disk" aperture, and 2–3 annuli tiling the known disk extent, in each case scaling the noise by $\sqrt{}$(number of independent beams) in the aperture.

- *Spectral stacking.* Stack the spectrum over each aperture at zero velocity shift (the plain stack): average the per-pixel spectra weighted by inverse variance, and measure the RMS of the stacked spectrum directly from its line-free channels rather than trusting the propagated value. At 49.6 km/s channels, galaxy rotation moves emission across only ~1–3 channels over most of the disk, so this plain stack captures nearly all the available sensitivity gain; quantify that expectation with a one-plot calculation (channel width versus plausible projected rotation spread) and include it.

- *Shift-and-stack (stretch goal only).* If — and only if — a published disk geometry for NGC 4429 (systemic velocity, PA, inclination, and a rotation-curve parameterization) is straightforwardly available in the literature, implement a velocity-coherent stack: shift each pixel's spectrum by the model projected velocity before averaging, and compare the resulting RMS to the plain stack. Timebox this to two weeks. If the geometry is not cleanly available, skip it; the thesis's limits are the plain-stack limits either way, and the write-up states which path was taken.

- *Reporting.* Every limit is reported as: assumed line width, aperture, 3σ flux limit in Jy km/s, and the corresponding mass limit from Section 7. The upper-limit table is the central result table of this branch.

5. Continuum measurement (either branch, ~2 weeks). On the wideband Band 3 continuum image: measure the RMS as in Section 1, then test for a nuclear source at the SMBH position. If a source is present at $\geq 5\sigma$, fit a 2-D Gaussian ('astropy.modeling' or 'photutils') to obtain position, peak and integrated flux density, and whether it is resolved (deconvolved size versus beam); compare the fitted position to the ICRS nuclear coordinates and report the offset with its uncertainty (~beam/(2×S/N)). If no source is present, report the 3σ point-source upper limit at the nuclear position. In either case, also run the same photometry over the full circumnuclear-disk aperture to constrain any extended 3 mm emission. Convert the nuclear continuum flux or limit to a monochromatic luminosity at the adopted Virgo distance (state the distance and its source explicitly; use it consistently everywhere). This number enters the discussion of AGN-related versus secular context exactly once, as a measured quantity.

6. Central-deficit constraint envelope (either branch, ~2–3 weeks, written as a short self-contained section). Reuse the Section 2 machinery to turn the central-beam measurement into a constraint on cavity parameters:

- Model grid: disk surface-brightness profiles of two functional forms (exponential and Gaussian) over a plausible range of scale lengths (bracket, don't fix — e.g., 0.5–3× the observed

HCN or literature CO disk scale), each with a central cavity of radius R_cav (grid from ~10 pc to ~1.5 beam FWHM in physical units) and fractional depth d (0–100%).

- For each model: normalize to the observed total flux (detection branch) or to the aperture limit (non-detection branch), convolve with the true beam, and compute the predicted central-beam flux and radial profile.
- Comparison: a model (R_cav, d) is excluded if its predicted central-beam flux differs from the observed value (or violates the limit) by $>3\sigma$, for *every* disk profile in the bracket. The reported exclusion region is this conservative intersection; any (R_cav, d) cell excluded under one profile assumption but not another is labeled unconstrained, not excluded.
- Deliverable: one two-panel figure — the (R_cav, d) exclusion map, and the observed radial profile overplotted with representative filled and holed models — plus one summary sentence of the form "cavities smaller than Y pc or shallower than Z% are unconstrained at this resolution; cavities of order the beam and near-total in depth are excluded/required." Fill in Y and Z from the grid; do not editorialize beyond what the grid shows.

7. Mass conversion and the bracketed mass budget (either branch, ~1 week). Convert the HCN(1-0) flux (or limit) to line luminosity L′_HCN in K km/s pc^2 via the standard Solomon & Vanden Bout relation, using the adopted distance and the observed (or assumed) line frequency. Convert to dense-gas mass with the standard α_HCN conversion factor, but treat α_HCN as a bracket, not a number: compute the mass for the canonical value and for the factor-of-several range spanned in the literature, and report every mass as a range with the conversion-factor systematic quoted separately from the statistical error. Do this for the central beam and for the full-disk aperture. If, and only if, a published CO flux for NGC 4429 can be defensibly matched to your HCN aperture — meaning you can establish, with real work, that the CO measurement's aperture and resolution are commensurable with yours (convolve/rescale as needed, document every step) — compute an HCN/CO dense-gas-fraction (or its limit) for the matched aperture. Budget two weeks for that matching; if it cannot be done defensibly, report the HCN mass (or limit) alone, which is the pre-committed fallback and a complete result.

8. Assembly and internal cross-checks (final month). Re-run the entire pipeline end-to-end from the two FITS files with a single driver script and confirm every thesis number regenerates. Cross-checks to perform and record: moment-0 total flux versus integrated-spectrum flux (must agree within errors); aperture photometry versus Gaussian-fit flux for the continuum; toy-model flux conservation under convolution; and the gate-memo numbers versus the final measured values. The thesis chapters map directly onto the sections above: data and verification (§1), gate (§3), the one executed branch (§4A or §4B), continuum (§5), cavity constraint envelope (§6), mass budget (§7), with the convolution sanity check (§2) as an appendix.